

AWS EBS Volume Management and Cross-Region Snapshot Migration

Introduction

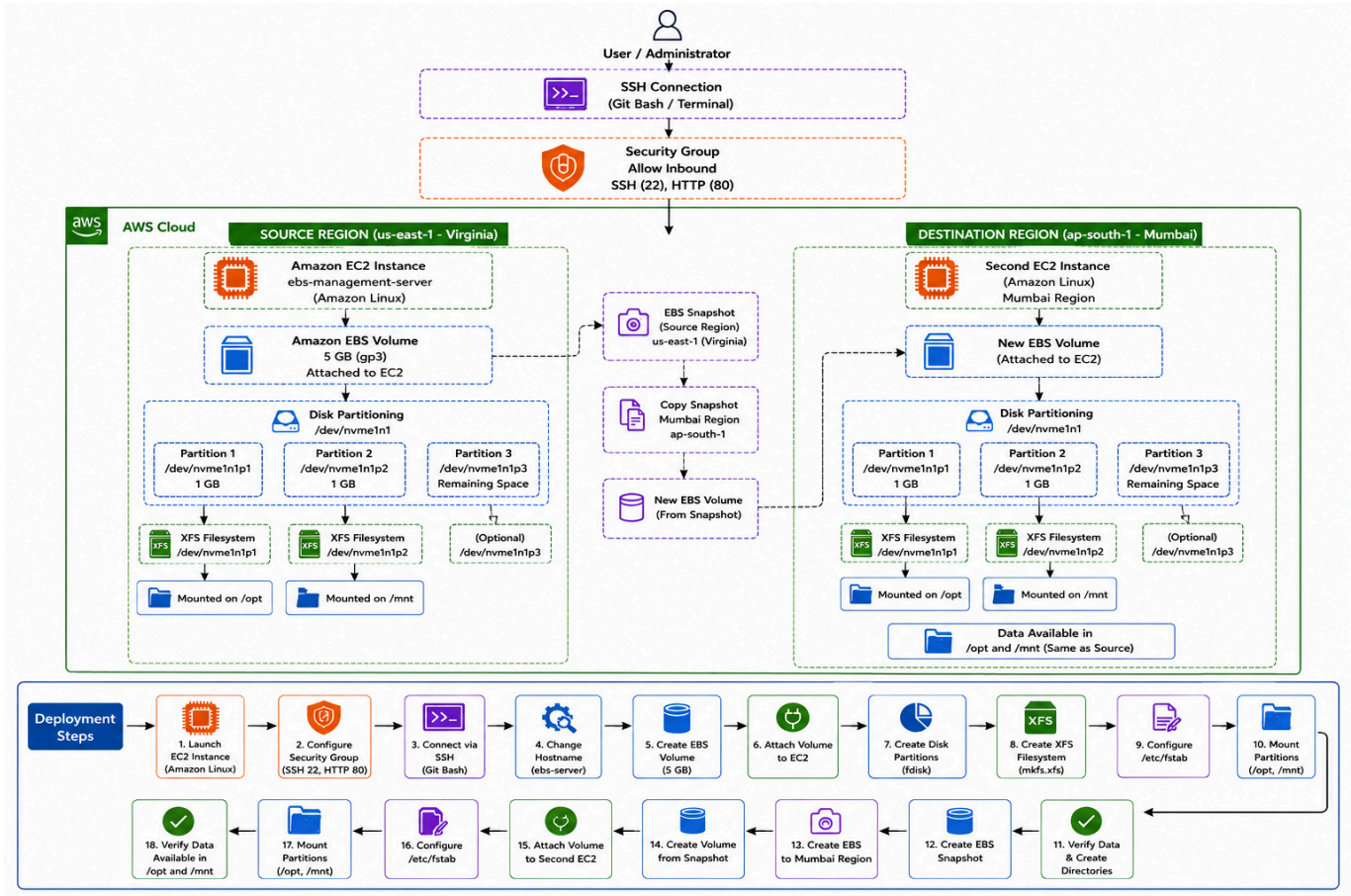
This project demonstrates practical implementation of AWS Elastic Block Store (EBS) management using Amazon EC2 instances. The project covers EBS volume creation, partitioning, filesystem creation, mounting, unmounting, snapshot creation, and cross-region migration using AWS services.

The main objective of this project is to understand persistent block storage management in AWS Cloud Infrastructure.

Services Used

- Amazon EC2
 - Amazon EBS
 - EBS Snapshot
 - Amazon Linux
 - SSH (Git Bash)
-

Architecture Diagram



EC2 Configuration

Configuration	Details
Instance Name	ebs-volume-demo-server
Operating System	Amazon Linux
Security Group	SSH and HTTP Enabled
Open Ports	22, 80
Hostname	ebs-storage-server

Project Workflow

Step 1 — Launch EC2 Instance

- Created an EC2 instance named `ebs-volume-demo-server`
- Used Amazon Linux AMI
- Allowed inbound rules for:
 - SSH (22)
 - HTTP (80)

EC2 instance

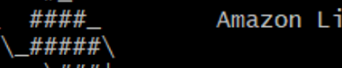
Security Group

Step 2 — Connect to EC2 Instance

```
ssh -i my-key.pem ec2-user@32.197.34.176
```

```
sudo hostnamectl set-hostname ebs-storage-server
```

Screenshot



```

#_
####_
#####\
#####|
\###|
\#/
V~'-'>
./
/m/'

```

Amazon Linux 2023

<https://aws.amazon.com/linux/amazon-linux-2023>

Volumes (1/2) Info

Saved filter sets

Choose filter set

Search

	Name	Volume ID	Type	Size	IOPS	Throughput
☐		vol-0943c40f55cb86a64	gp3	5 GiB	3000	125
☒		vol-0fe89a977c000e846	gp3	8 GiB	3000	125

Volume ID: vol-0fe89a977c000e846

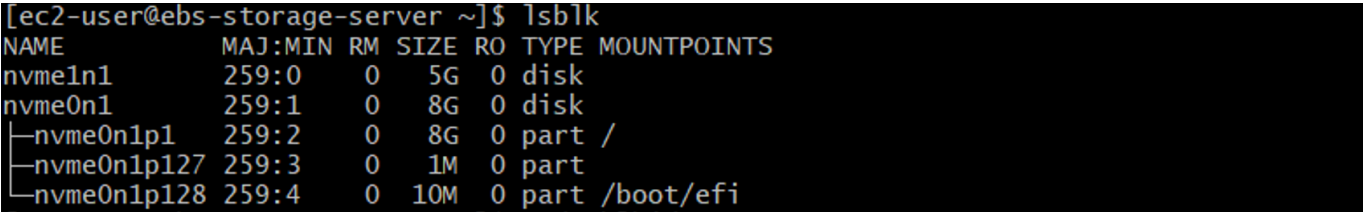
Details	Status checks	Monitoring	Tags
Volume ID vol-0fe89a977c000e846 AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more Fast snapshot restored No		Size 8 GiB Volume state In-use Availability Zone aps1-az1 (ap-south-1a)	Type gp3 IOPS 3000 Created Sun Standard

Step 5 — Verify Attached Volume

Checked attached volume using:

```
lsblk
```

Screenshot



Step 6 — Create Partitions

Used `fdisk` to create 3 partitions.

```
sudo fdisk /dev/nvme1n1
```

Partition sizes:

- Partition 1 → 1GB
- Partition 2 → 1GB

Screenshot

```
[ec2-user@ebs-storage-server dev]$ sudo fdisk nvme1n1

welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x7f7cbc8a.

Command (m for help): n
Partition type
   p   primary (0 primary, 0 extended, 4 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-10485759, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-10485759, default 10485759): +1G

Created a new partition 1 of type 'Linux' and of size 1 GiB.

Command (m for help): n
Partition type
   p   primary (1 primary, 0 extended, 3 free)
   e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (2-4, default 2):
First sector (2099200-10485759, default 2099200):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2099200-10485759, default 10485759): +1G

Created a new partition 2 of type 'Linux' and of size 1 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks
```

Step 7 — Create Filesystem

Created XFS filesystem:

```
sudo mkfs.xfs /dev/nvme1n1p1
sudo mkfs.xfs /dev/nvme1n1p2
```

Verified using:

```
sudo blkid
```

Screenshot

```
[ec2-user@ebs-storage-server dev]$ blkid
/dev/nvme0n1p1: LABEL="/" UUID="afcd3653-4916-4996-a80d-b02b83e8be92" BLOCK_SIZE="4096" TYPE="xfs" PARTLABEL="Linux" PARTUUID="4cfd0ea4-568c-46d4-9671-698222961497"
/dev/nvme0n1p128: SEC_TYPE="msdos" UUID="7580-447F" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI System Partition" PARTUUID="5a0ab02b-6aae-4454-aaeb-1176ccc669cb"
[ec2-user@ebs-storage-server dev]$ sudo blkid
/dev/nvme0n1p1: LABEL="/" UUID="afcd3653-4916-4996-a80d-b02b83e8be92" BLOCK_SIZE="4096" TYPE="xfs" PARTLABEL="Linux" PARTUUID="4cfd0ea4-568c-46d4-9671-698222961497"
/dev/nvme0n1p128: SEC_TYPE="msdos" UUID="7580-447F" BLOCK_SIZE="512" TYPE="vfat" PARTLABEL="EFI System Partition" PARTUUID="5a0ab02b-6aae-4454-aaeb-1176ccc669cb"
/dev/nvme0n1p127: PARTLABEL="BIOS Boot Partition" PARTUUID="8d489d5b-35d9-4bd4-8f47-0c3fb274f36d"
/dev/nvme1n1p2: UUID="6db104bd-00ea-4e36-b0c2-be48e911dea9" BLOCK_SIZE="512" TYPE="xfs" PARTUUID="7f7cbc8a-02"
/dev/nvme1n1p1: UUID="89c4a952-22c0-4194-b4dc-0d035dfcb0df" BLOCK_SIZE="512" TYPE="xfs" PARTUUID="7f7cbc8a-01"
[ec2-user@ebs-storage-server dev]$
```

Edited /etc/fstab

Added:

Screenshot

Step 9 — Mount Filesystems

Mounted partitions:

```
sudo mount /opt
sudo mount /mnt
```

Verified:

```
lsblk
```

Screenshot

```
[ec2-user@ebs-storage-server mnt]$ sudo vim /etc/fstab
[ec2-user@ebs-storage-server mnt]$ sudo mount /opt
[ec2-user@ebs-storage-server mnt]$ sudo mount /mnt
[ec2-user@ebs-storage-server mnt]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme1n1              259:0    0   5G  0 disk
├─nvme1n1p1          259:7    0    1G  0 part /opt
├─nvme1n1p2          259:8    0    1G  0 part /mnt
nvme0n1              259:1    0   8G  0 disk
├─nvme0n1p1          259:2    0    8G  0 part /
├─nvme0n1p127        259:3    0    1M  0 part
└─nvme0n1p128        259:4    0   10M  0 part /boot/efi
[ec2-user@ebs-storage-server mnt]$
```

Step 10 — Verify Data

Created test directories:

```
cd /opt
mkdir disk-2

cd /mnt
mkdir part-2
```

Screenshot

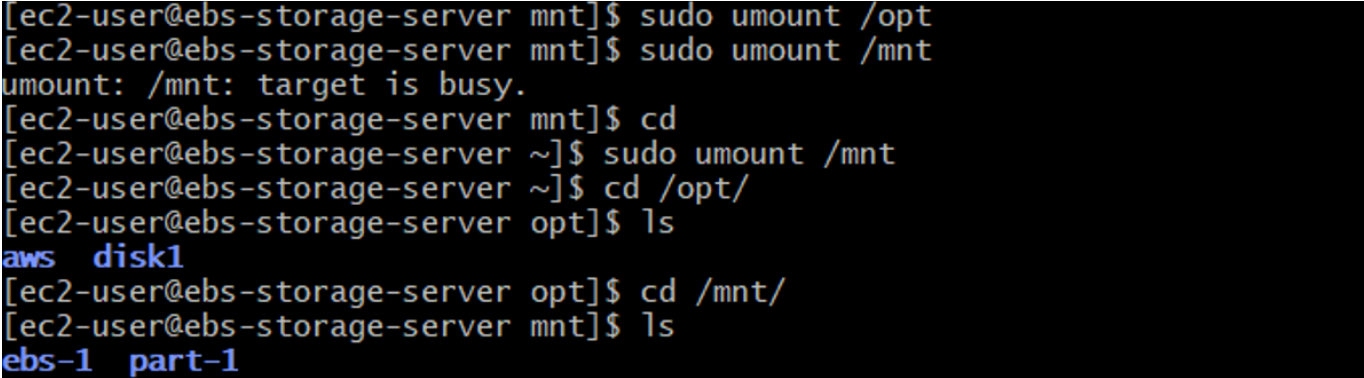
```
[ec2-user@ebs-storage-server mnt]$ sudo mount /opt/
[ec2-user@ebs-storage-server mnt]$ sudo mount /mnt/
[ec2-user@ebs-storage-server mnt]$ cd /opt/
[ec2-user@ebs-storage-server opt]$ ls
disk-2
[ec2-user@ebs-storage-server opt]$ cd /mnt/
[ec2-user@ebs-storage-server mnt]$ ls
part-2
```

Step 11 — Unmount Filesystems

Unmounted partitions:

```
sudo umount /opt
sudo umount /mnt
```

Screenshot

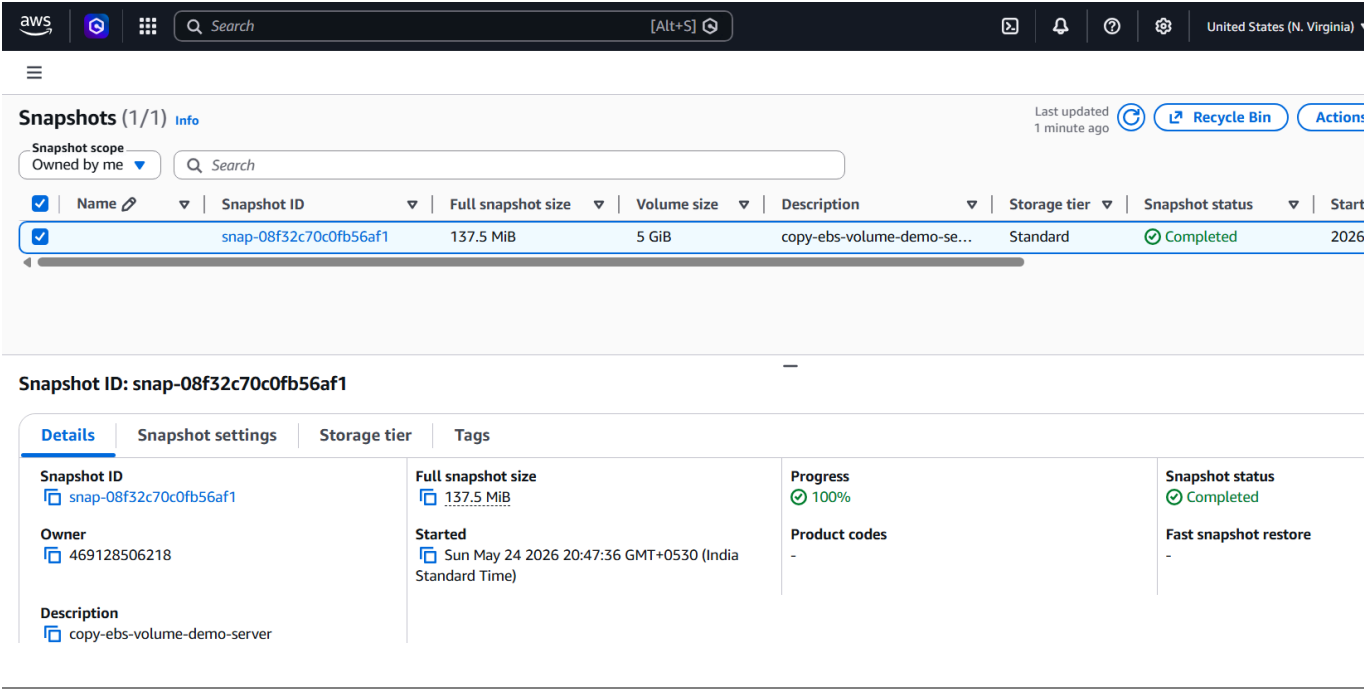


Cross Region Snapshot Migration

Step 12 — Create Snapshot

Created snapshot of EBS volume in source region.

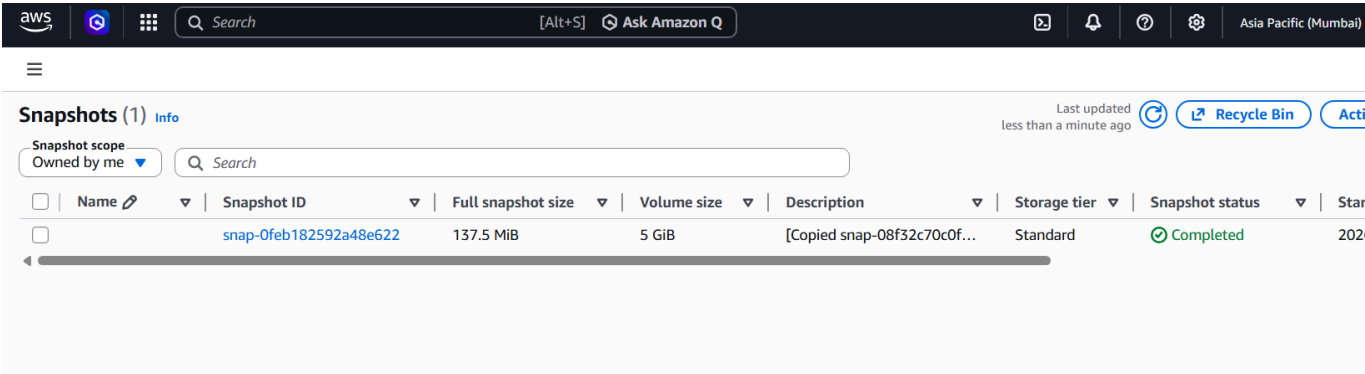
Screenshot



Step 13 — Copy Snapshot to Mumbai Region

Copied snapshot to another AWS region.

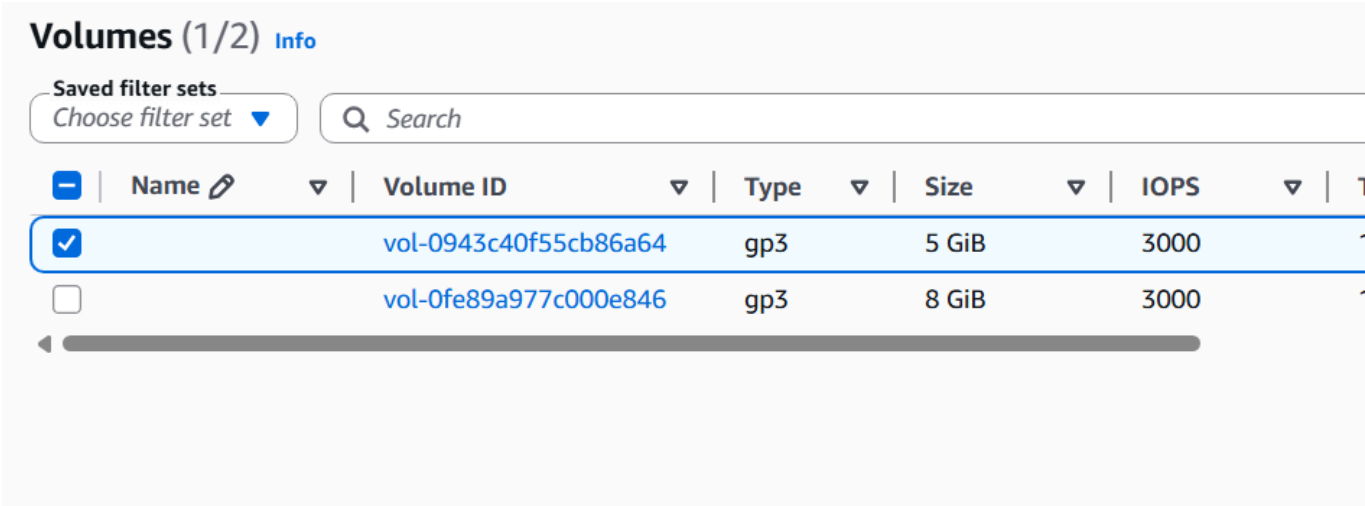
Screenshot



Step 14 — Create Volume from Snapshot

Created new EBS volume from copied snapshot.

Screenshot



Volume ID: vol-0943c40f55cb86a64

Details	Status checks	Monitoring	Tags
Volume ID  vol-0943c40f55cb86a64			
AWS Compute Optimizer finding  Opt-in to AWS Compute Optimizer for recommen- dations. Learn more 			
Fast snapshot restored No			
	Size  5 GiB		
	Volume state  Available		
	Availability Zone aps1-az1 (ap-south-1a)		

Step 15 — Attach Volume to New EC2 Instance

Attached volume to new EC2 instance and mounted partitions.

Screenshot

Volumes (1/2) Info

Last updated 2 minutes ago

Recycle Bin

Actions

Save filter sets

Choose filter set

Search

	Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume
☒		vol-0943c40f55cb86a64	gp3	5 GiB	3000	125	snap-0feb182...	-
☐		vol-0fe89a977c000e846	gp3	8 GiB	3000	125	snap-0bb75e9...	-

Modify volume

Create snapshot

Create snapshot lifecycle policy

Delete volume

Attach volume

Detach volume

Force detach volume

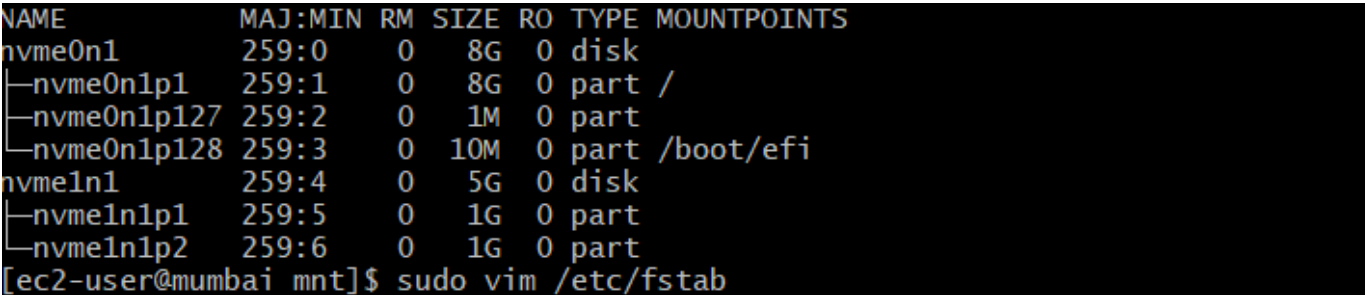
Manage auto-enabled I/O

Step 16 — Verify Attached Volume

Checked attached volume using:

```
lsblk
```

Screenshot



Step 17 — Configure Mount Points

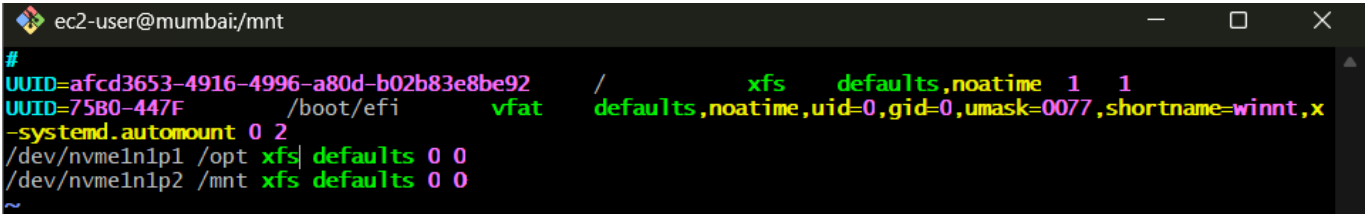
Edited `/etc/fstab`

```
sudo vim /etc/fstab
```

Added:

```
/dev/nvme1n1p1 /opt xfs defaults 0 0
/dev/nvme1n1p2 /mnt xfs defaults 0 0
```

Screenshot



Step 18 — Mount directories

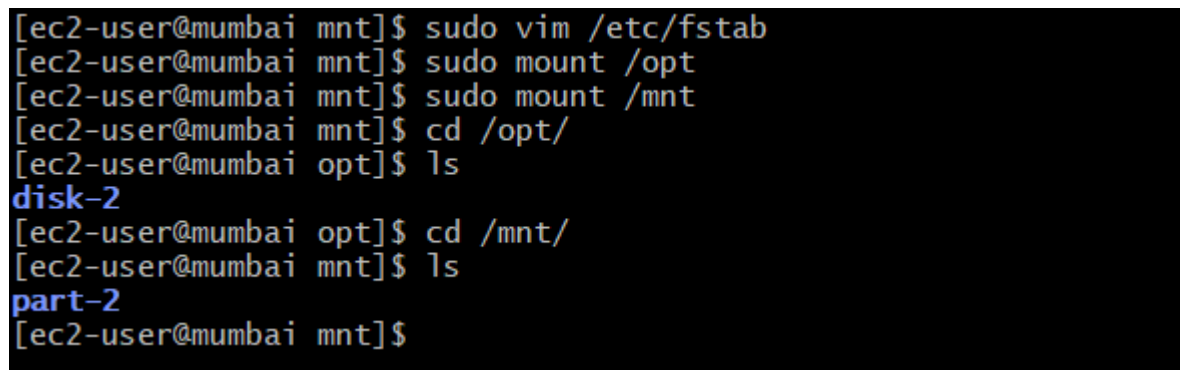
Mounted partitions:

```
sudo mount /opt
sudo mount /mnt
```

Verified:

```
lsblk
```

Screenshot



```
[ec2-user@mumbai mnt]$ sudo vim /etc/fstab
[ec2-user@mumbai mnt]$ sudo mount /opt
[ec2-user@mumbai mnt]$ sudo mount /mnt
[ec2-user@mumbai mnt]$ cd /opt/
[ec2-user@mumbai opt]$ ls
disk-2
[ec2-user@mumbai opt]$ cd /mnt/
[ec2-user@mumbai mnt]$ ls
part-2
[ec2-user@mumbai mnt]$
```

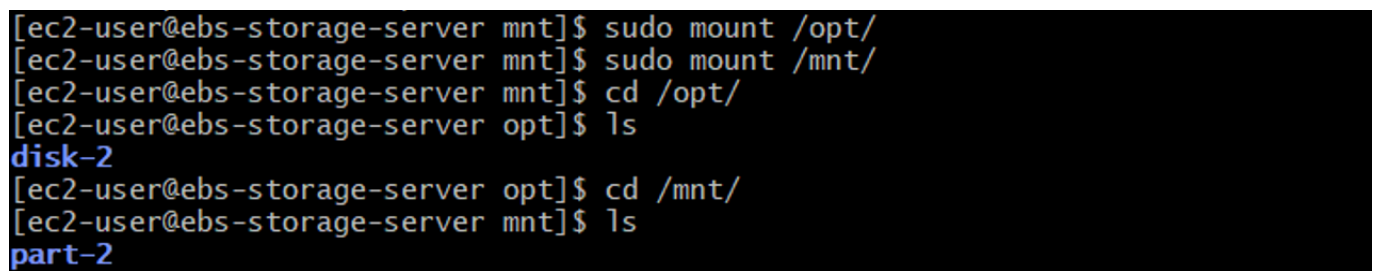
Step 19 — Verify Data

Created test directories:

```
cd /opt
ls

cd /mnt
ls
```

Screenshot



```
[ec2-user@ebs-storage-server mnt]$ sudo mount /opt/
[ec2-user@ebs-storage-server mnt]$ sudo mount /mnt/
[ec2-user@ebs-storage-server mnt]$ cd /opt/
[ec2-user@ebs-storage-server opt]$ ls
disk-2
[ec2-user@ebs-storage-server opt]$ cd /mnt/
[ec2-user@ebs-storage-server mnt]$ ls
part-2
```

Step 20 — Verify Data in New Region

Verified data persistence inside:

- /opt
- /mnt

Screenshot

before mount

```
ec2-user@mumbai:/mnt
[ec2-user@mumbai ~]$ lsblk
NAME                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
nvme0n1              259:0    0   8G  0 disk
├─nvme0n1p1          259:1    0   8G  0 part /
├─nvme0n1p127        259:2    0   1M  0 part
└─nvme0n1p128        259:3    0  10M  0 part /boot/efi
[ec2-user@mumbai ~]$ cd /opt/
[ec2-user@mumbai opt]$ ls
aws
[ec2-user@mumbai opt]$ cd /mnt/
[ec2-user@mumbai mnt]$ ls
[ec2-user@mumbai mnt]$
```

after mount

```
[ec2-user@mumbai mnt]$ sudo mount /opt
[ec2-user@mumbai mnt]$ sudo mount /mnt
[ec2-user@mumbai mnt]$ cd /opt/
[ec2-user@mumbai opt]$ ls
disk-2
[ec2-user@mumbai opt]$ cd /mnt/
[ec2-user@mumbai mnt]$ ls
part-2
[ec2-user@mumbai mnt]$
```

Commands Used

```
lsblk
sudo fdisk /dev/nvme1n1
sudo mkfs.xfs /dev/nvme1n1p1
sudo mkfs.xfs /dev/nvme1n1p2
sudo blkid
sudo vim /etc/fstab
sudo mount /opt
sudo mount /mnt
sudo umount /opt
sudo umount /mnt
```

Key Learnings

- EBS volume management
 - Disk partitioning in Linux
 - Filesystem creation using XFS
 - Persistent mounting using `/etc/fstab`
 - Snapshot backup process
 - Cross-region snapshot migration
 - Data persistence across EC2 instances
-

Conclusion

This project successfully demonstrated AWS EBS volume management and cross-region migration using snapshots. The implementation provided practical understanding of cloud storage administration, Linux disk management, filesystem handling, and disaster recovery concepts in AWS Cloud.

Author

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